Name: Praju Maharjan

Group: L5CG9

Tutorial - 6

1. Decode a PNG image into 1D and 2D array image.

Ans :

Source Code :

```
#include <stdio.h>
#include <stdlib.h>
#include "lodepng.h"

int main(){
 unsigned char *image;
 unsigned width,height;
 unsigned int error;

 error=lodepng_decode32_file(&image,&width,&height,"4x4_Image.png");
 if(error){
  printf("Error : %d,Message: %s",error,lodepng_error_text(error));
  return 1;
 }
 printf("Image loaded\n");

 printf("\n\nImage decoded into 1D array image : \n");
 for(int i=0;i<width*height*4;i+=4){
  printf("%d %d %d %d \n",image[i],image[i+1],image[i+2],image[i+3]);
 }

 unsigned char image2D[height][width*4];

 for(int i=0;i<height;i++){
  for(int j=0;j<width;j++){
   int index=(i*width+j)*4;
   int n_col=j*4;

   image2D[i][n_col]=image[index];
   image2D[i][n_col+1]=image[index+1];
   image2D[i][n_col+2]=image[index+2];
   image2D[i][n_col+3]=image[index+3];
  }
 }

 printf("\n\nImage decoded into 2D array image : \n");
 for(int i=0;i<height;i++){
  for(int j=0;j<width;j++){
    int n_col=j*4;
    printf("%d %d %d %d|",image2D[i][n_col],image2D[i][n_col+1],image2D[i][n_col+2],image2D[i][n_col+3]);
  }
  printf("\n");
 }
 free(image);
 printf("\n");
 return 0;
}
```

Output :

```
Image loaded


Image decoded into 1D array image :
40 23 255 255
40 23 255 255
255 36 36 255
255 36 36 255
40 23 255 255
40 23 255 255
255 36 36 255
255 36 36 255
255 203 23 255
255 203 23 255
34 177 76 255
34 177 76 255
255 203 23 255
255 203 23 255
34 177 76 255
34 177 76 255


Image decoded into 2D array image :
40 23 255 255|40 23 255 255|255 36 36 255|255 36 36 255|
40 23 255 255|40 23 255 255|255 36 36 255|255 36 36 255|
255 203 23 255|255 203 23 255|34 177 76 255|34 177 76 255|
255 203 23 255|255 203 23 255|34 177 76 255|34 177 76 255|
```

2. Apply a negative filter on an image file. This is the process of reversing the RGB values. For example, if the Red value of the pixel is 100, the inverse is 155. NOTE: 0 is minimum and 255 is maximum.

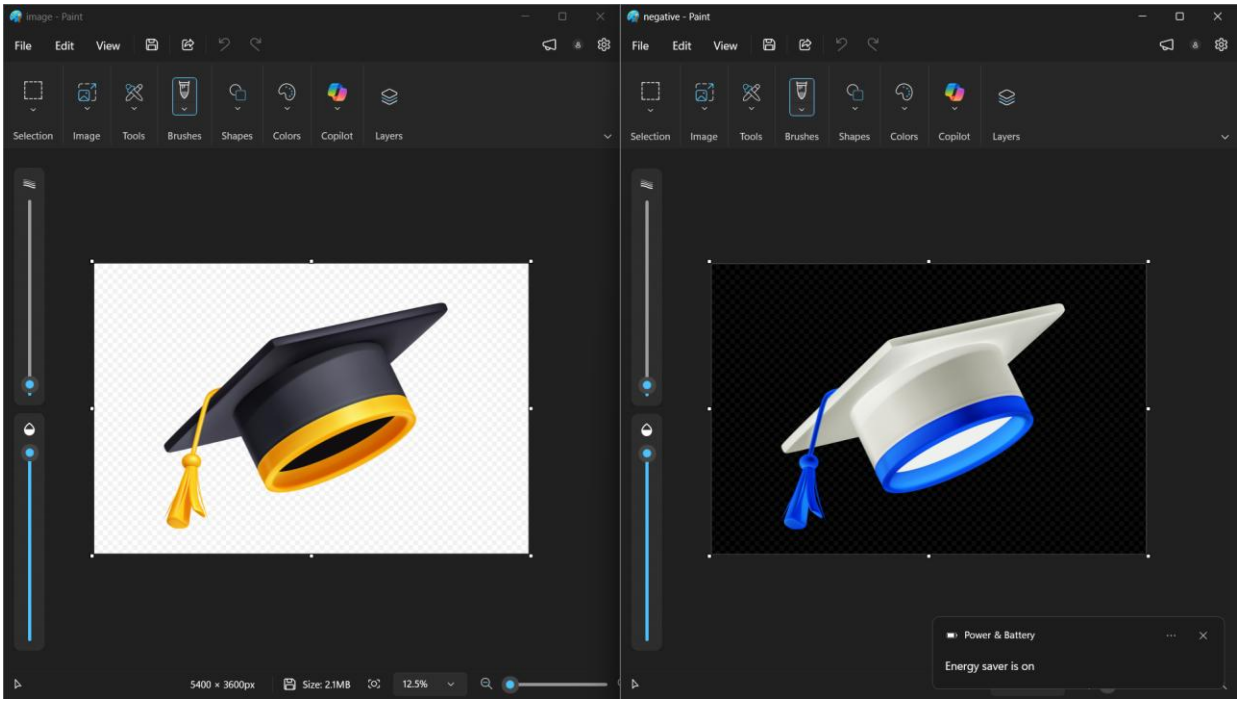
Ans :

Source Code :

```c
#include <stdio.h>
#include <stdlib.h>
#include "lodepng.h"

int main(){
    unsigned char *image;
    unsigned width, height;
    unsigned error;

    error=lodepng_decode32_file(&image,&width,&height,"4x4_Image.png");
    if(error){
        printf("Decode error %u: %s\n", error, lodepng_error_text(error));
        return 1;
    }
    for(unsigned i=0;i<width*height*4;i+=4){
        image[i]=255-image[i];
        image[i+1]=255-image[i+1];
        image[i+2]=255-image[i+2];
    }

    error=lodepng_encode32_file("negative.png",image,width,height);
    if(error){
        printf("Encode error %u: %s\n", error, lodepng_error_text(error));
        free(image);
        return 1;
    }

    printf("Negative image saved as negative.png\n");

    free(image);
    printf("\n");
    return 0;
}
```

Output :

```
Negative image saved as negative.png
```



3. Apply a grayscale filter on an image file. Please refer lecture slide for the grayscale formula.

Ans :

Source Code :

```c
#include <stdio.h>
#include <stdlib.h>
#include "lodepng.h"

int main(){
        unsigned char *image;
        unsigned width,height;
        unsigned error;

        error=lodepng_decode32_file(&image,&width,&height,"image.png");
        if (error){
                printf("Decode Error %u : %s \n",error,lodepng_error_text(error));
                return 1;
        }

        for(unsigned i=0;i<width*height*4;i+=4){
                unsigned char gray=(unsigned char)(0.299*image[i]+0.587*image[i+1]+0.113-image[i+2]);

                image[i]=gray;
                image[i+1]=gray;
                image[i+2]=gray;
        }

        error=lodepng_encode32_file("grayscaled_image.png",image,width,height);
        if(error){
                printf("Encode Error %u : %s\n",error,lodepng_error_text(error));
                free(image);
                return 1;
        }

        printf("Image turned to grayscale successfully");
        free(image);
        printf("\n");
        return 0;
}
```

Output :

```
Image turned to grayscale successfully
```

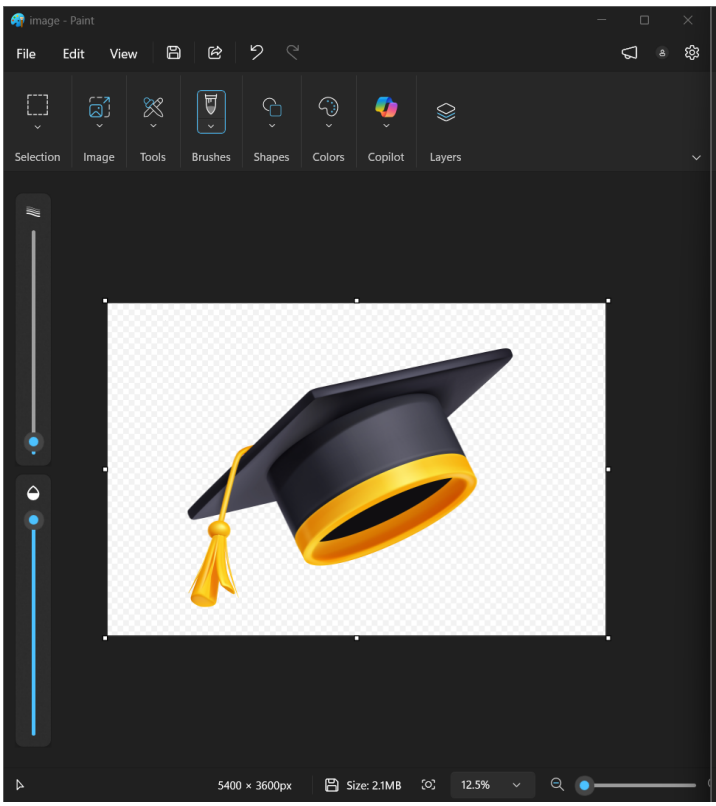
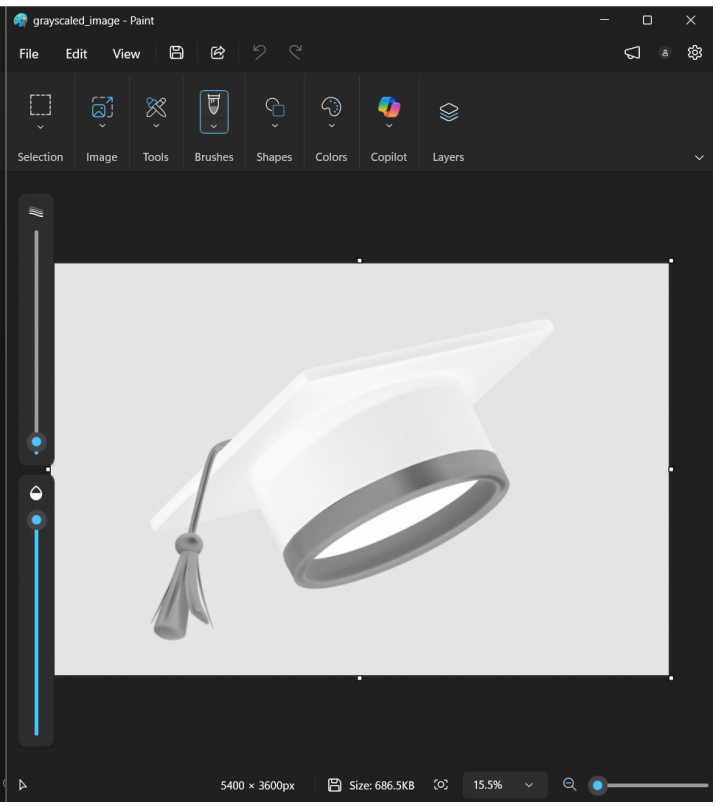


image - Paint
File
Edit
View
Selection
Image
Tools
Brushes
Shapes
Colors
Copilot
Layers
5400 × 3600px
Size: 2.1MB
12.5%


grayscaled_image - Paint
File
Edit
View
Selection
Image
Tools
Brushes
Shapes
Colors
Copilot
Layers
5400 × 3600px
Size: 686.5KB
15.5%